

Problem Set 3, COSC330/PHYS306

Chapters 3 and 4

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1 Chapter 3

1. [Section 3-9, Exercise 39] Examine the conditions indicated in Figure 3-92, and identify the faulty gates.

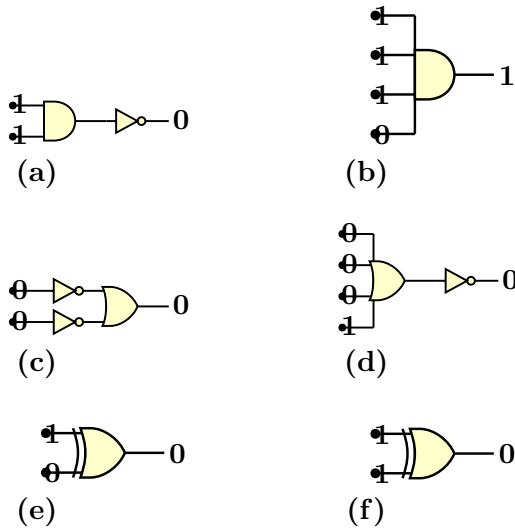


FIGURE 3-92

2. [Section 3-9, Exercise 40] Determine the faulty gates in Figure 3-93 by analyzing the timing diagrams.

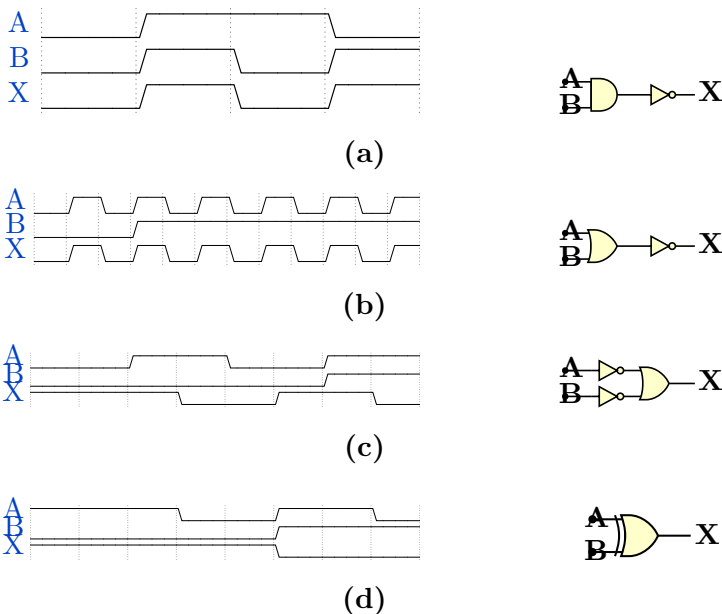


FIGURE 3-93

2 Chapter 4

3. [Section 4-3, Exercise 9] Apply DeMorgan's theorems to each expression:

- (a) $\overline{A + B}$
- (b) $\overline{\overline{AB}}$
- (c) $\overline{A + B + C}$
- (d) $\overline{ABC}$
- (e) $\overline{A(B + C)}$
- (f) $\overline{AB + CD}$
- (g) $\overline{\overline{AB} + CD}$
- (h) $\overline{(A + \overline{B})(\overline{C} + D)}$

4. [Section 4-4, Exercise 13] Write the Boolean expression for each of the logic circuits in Figure 4-57.

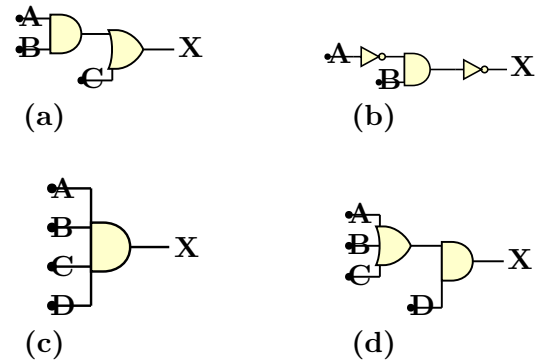


FIGURE 4-57

5. [Section 4-4, Exercise 15] Draw the logic circuit represented by each expression:

- (a) $A(B + C)$
- (b) $AB + ABC$
- (c) $\overline{ABC} + D$
- (d) $A + B(C + DB + \overline{C})$

6. [Section 4-4, Exercise 18] Construct a truth table for each of the following Boolean expressions:

- (a) $A + B$
- (b) AB

(c) $A + B(C + D)$

(d) $A + B + C$

(e) $AB + BC$

7. [Section 4-4, Exercise 22] Determine which of the logic circuits in Figure 4-59 are equivalent.

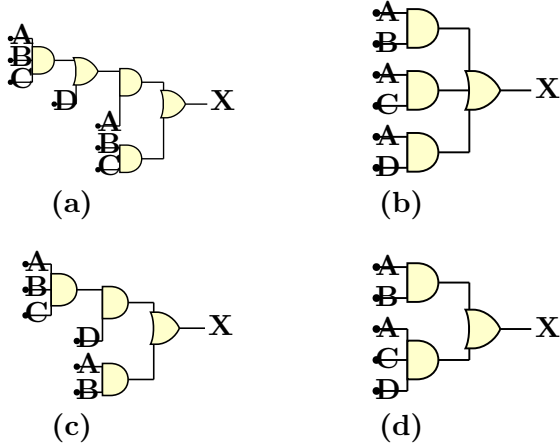


FIGURE 4-59

8. [Section 4-7, Exercise 31] Develop a truth table for each of the following standard SOP expressions:

(a) $ABC + \overline{A}BC + ABC\overline{C}$

(b) $XYZ + X\overline{Y}Z + \overline{X}YZ + \overline{X}\overline{Y}\overline{Z} + X\overline{Y}\overline{Z}$

9. [Section 4-7, Exercise 36] For each truth table in Table 4-15, derive a standard SOP and a standard POS expression.

ABC	X	ABC	X
0 0 0	0	0 0 0	0
0 0 1	1	0 0 1	0
0 1 0	0	0 1 0	0
0 1 1	0	0 1 1	0
1 0 0	1	1 0 0	0
1 0 1	1	1 0 1	1
1 1 0	0	1 1 0	1
1 1 1	1	1 1 1	1

(a)

(b)

$ABCD$	X	$ABCD$	X
0 0 0 0	1	0 0 0 0	0
0 0 0 1	1	0 0 0 1	0
0 0 1 0	0	0 0 1 0	1
0 0 1 1	1	0 0 1 1	0
0 1 0 0	0	0 1 0 0	1
0 1 0 1	1	0 1 0 1	1
0 1 1 0	1	0 1 1 0	0
0 1 1 1	0	0 1 1 1	1
1 0 0 0	0	1 0 0 0	0
1 0 0 1	1	1 0 0 1	0
1 0 1 0	0	1 0 1 0	0
1 0 1 1	0	1 0 1 1	1
1 1 0 0	1	1 1 0 0	1
1 1 0 1	0	1 1 0 1	0
1 1 1 0	0	1 1 1 0	0
1 1 1 1	0	1 1 1 1	1

(c)

(d)

TABLE 4-15

10. [Section 4-9, Exercise 40] Use a Karnaugh map to find the minimum SOP form for each expression:

(a) $\overline{A}\overline{B}C + \overline{A}BC + \overline{A}B\overline{C}$

(b) $AC(\overline{B} + C)$

(c) $\overline{A}(BC + B\overline{C}) + A(BC + B\overline{C})$

(d) $\overline{A}\overline{B}C + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$